

々 To create something from a file

kubectl create -f <file path> [it'll create if resource doesn't exist]

kubectl apply -f <file path> [create if doesn't exist; update if exists]

々 To edit any resource live

kubectl edit <resource type> <resource name>

kubectl edit pod my-pod

々 To see all the details of a resource along with events

kubectl describe <resource type> <resource name>

kubectl describe pod my-pod

々 To see *live resource configuration* in *yaml* format

kubectl get <resource type> <resource name> -o yaml

kubectl get pod my-pod -o yaml

々 Too see all the things of the available resources

kubectl get <resource type> <resource name> -o wide

々 To get a *yaml* configuration file of any resource to start with,

--dry-run=client -o yaml append this in the imperative commands.

kubectl create deployment my-dep --image=nginx --dry-run=client -o yaml

々 Pod

♣ To create a pod ("v1")

kubectl run <pod name> --image=<image name>

kubectl run my-pod --image=nginx

♣ Whatever ports you write inside containers (inside pods) are not exposed, that is just used for user's docs or to search pods a/c to ports. The traffic reaches to a specific port of a container with Service only.

♣ If a pod contains multiple containers,
each container: *same IP, different ports* [because IP is assigned to Pod]

々 ReplicaSet

♣ ReplicaSet ("*apps/v1*") and ReplicationController ("*v1*") cannot be created with imperative commands.

♣ ReplicaSet contains [*replicaset.spec.selector*](#) field, which ReplicationController doesn't have.

♣ Selector contains [*matchLabels*](#) or [*matchExpressions*](#).

- ⌘ *template* (pod) should contain same or more labels present in *selector*.
Selector's labels work with AND rule, not OR. All the labels must be present in the pods to be selected.

- ⌘ To scale replica set:

kubectl scale --replicas=<replica count> replicaset <replicaset name>

kubectl scale --replicas=4 replicaset my-rs

✧ Deployment

- ⌘ Deployment can be created with Imperative way because Kubernetes auto-fill the required complex fields like template, labels, selectors.

kubectl create deployment <deployment name> --image=<image name> --replicas=<replica count>

- ⌘ It provide *pull*, *upgrade*, *rolling update (or) recreate*, *rollback (undo)*, *pause (rollout pause)*, *resume (rollout resume)* over ReplicaSet.
- ⌘ It create new versions, and *scales down* old ReplicaSet gradually and *scales up* new ReplicaSet gradually.

- ⌘ maxUnavailable (maximum allowed unavailable pods. Percentage or count), maxSurge (maximum allowed available pods. Percentage or count)

[scale up of new pods & scale down of old pods happens w.r.t.

maxUnavailable & *maxSurge*]

- ⌘ To see *rollout status*,

kubectl rollout status deployment <deployment name>

To see *rollout history*,

kubectl rollout history deployment <deployment name>

[for this --record has to be provided; but its deprecated]

To *undo* the deployment

kubectl rollout undo deployment <deployment name>

- ⌘ Deployment = ReplicaSet + [rollout, rollback, versioning(revisions), rollout strategy (Recreate, RollingUpdate), Pause, Resume]

✧ Deployment Strategy

- ⌘ RollingUpdate: gradually remove old pods and create new pods.
- ⌘ Recreate: remove all old pods and create new pods.

✧ Service

- ⌘ Required to communicate with the pods.
[Pod's IP can be used; but shouldn't because Pods might get re-created]
- ⌘ External to Pod: **NodePort**
Pod to Pod: **ClusterIP** (NodePort can also be used; but not preferred)
Load Balancing: **LoadBalancer** [works in cloud based clusters]
- ⌘ type: NodePort
ports: [{ nodePort: x, port: y, targetPort: z }, { .. }, ..]
port means service's port
in here, whatever port you assign for *service*, it doesn't matter because it is there only to connect nodePort to targetPort (pod's port).
- ⌘ type: ClusterIP
ports: [{ port: x, targetPort: y }, { .. }, ..]
- ⌘ For inter-pod communication
<service name>:<service port>
service -> name, selector; pod: labels; <-- these are important
- ⌘ *selector* doesn't contain *matchLabels*, *matchExpressions*.
Direct *key:value* pairs (labels) to select the target pods.
- ⌘ Service's redirection to Pod is cluster level. So service's and pod's host nodes might be different.
- 々 **kubeadm** tool can be used to create a cluster and configure multiple nodes (control plane & worker nodes). it generates the required certificates and all.
<https://github.com/kodekloudhub/certified-kubernetes-administrator-course/tree/master/kubeadm-clusters/generic>
- 々 **cgroup** (Control Group)
 - ⌘ Used to enforce limits by Kernel.
 - ⌘ **mount | grep cgroup** : to see the mounted path (/sys/fs/cgroup)
cgroup2 for modern systems; cgroup for older systems;
 - ⌘ Files & Directories inside *cgroup* are handled by ?
 - ⌘ all files : *Kernel*
 - ⌘ directories of format *cgroup.**, *cpu.**, *memory.**, *io.** : *Kernel*
 - ⌘ directories of format *.slice*, *.scope*, *.service*, *.mount* : **systemd**
 - ⌘ other directories (*docker*, *kubepods* ..etc): respective owners

- ♣ *system.slice* contains *containerd.service*, it means container runtime (containerd) is managed by *systemd* itself.

In case of separately handled, both will not get to know each other's usage and request for more resources; at the end kernel will take action and stop some resources which might be important.

- ♣ Owners writes into the respective files, Kernel enforces limits.

✧ Internals of *kubectl apply*

- ♣ 3 configs are there:

config file (static yaml configuration file used to create a resource)

live config file (configuration file of the running resource; *kubectl get <resource type> <resource name> -o yaml*)

last applied config (latest config applied in *declarative* way) [its in JSON format]

- ♣ Running resource updated with:

- ♣ imperative command (or) *kubectl edit* command: live config is updated, last applied config will remain same
- ♣ Declarative way (*kubectl apply*) : live applied config will be updated, config file will be updated of-course.

- ♣ Configs updation:

If a field is there in *live config* (ex: an extra level was added using *kubectl edit* command) but not in both *config file* and *last applied config* then field will remain as it is.

If a field is there in either *config file* or *last applied config* or *both* then that field will be updated (doesn't matter if that field is there in *live config* or not)

Field in object configuration file	Field in live object configuration	Field in last-applied-configuration	Action
Yes	Yes	-	Set live to configuration file value.
Yes	No	-	Set live to local configuration.
No	-	Yes	Clear from live configuration.
No	-	No	Do nothing. Keep live value.

✧ namespace

- ♣ Namespace defines a logical boundary where resources can be present.

- Same named resources cannot stay inside a single namespace; but can be present in different namespaces.
- Ex: dev, prod, ..etc namespaces can be created.
- To query/update resources in a specific namespace, use
`-n <namespace name>` (or) `--namespace <namespace name>`
`kubectl get pods -n dev-namespace`
- In the yaml, `metadata.namespace` is the required field to set the namespace.
- Clustered configured with `kubeadm`, contains `kube-system` namespace which contains control plane resources.
- To access a service of another namespace
`<service name>.<target namespace name>.svc.cluster.local`
`db-service.dev-namespace.svc.cluster.local`

ResourceQuota

- Used to limit resources for **namespace** level.

```
spec:
  hard:
    pods: "10"
    requests.cpu: "4"
    requests.memory: 5Gi
    limits.cpu: "10"
    limits.memory: 10Gi
```

Scheduling

- `Scheduler` updates `pod.spec.nodeName` field of Pod.
 Then request API server to bind Pod to the specific node.
 Binding request contains payload, where `kind: bind`

```
apiVersion: v1
kind: Binding
metadata:
  name: test-binding
target:
  apiVersion: v1
  kind: Node
  name: node01
```

[not a Kubernetes object; just payload of request]

- `curl <request uri> --request POST --data '{"apiVersion": "v1", ...}'`
 (bind config in JSON format)

- ⌘ In case of manual scheduling, mention [*pod.spec.nodeName*](#) field and it'll create pod in required node.
- 々 To query resources with specific labels,
 - selector key1=val1,key2=val2** [append this]
 - `kubectl get all --selector app=nginx,key2=val2`
- 々 [*metadata.annotations*](#) used for giving extra info like build version info, adding tool details ..etc without impacting anything.
- 々 Taints & Tolerations
 - ⌘ Taint is applied on Node; Toleration is applied on Pod.
 - ⌘ Taint prevent pod to be scheduled to the tainted node;
 - If Pod is tolerated against that specific taint then only it can be scheduled to that tainted node.
 - ⌘ To taint node (*imperative method only; no declarative method*)
 - kubectl taint node <node name> <key>=<value>:<taint effect>**
 - `kubectl taint node node01 env=prod:NoSchedule`
 - ⌘ To remove taint, just append **-** at the end.
 - kubectl taint node <node name> <key>=<value>:<taint effect>-**
 - `kubectl taint node node01 env=prod:NoSchedule-` (most specific)
 - `kubectl taint node node01 env=prod-` (less specific)
 - `kubectl taint node node01 env-` (more less specific)
 - ⌘ One node can have more than one same key-value pair taints; but *taint effect* should be different.
 - `env=prod:NoSchedule`
 - `env=prod:NoExecute`
 - `env=prod:PreferNoSchedule`
 - ⌘ To tolerate the pod (*declarative method only; no imperative method*)
 - update the field [*pod.spec.tolerations*](#) [it contains a list]

```
spec:
  tolerations:
  - key: "env"
    operator: "Equal"
    value: "prod"
    effect: "NoSchedule"
```

 - operators: *Equal, Exists, Gt, Lt*
 - double quote is mandatory here.
 - ⌘ Taint effects:

- [NoSchedule](#): all current pods will present; non-tolerated pod will not be scheduled.
- [PreferNoSchedule](#): all current pods will present; non-tolerated pod might be scheduled in case of no other option.
- [NoExecute](#): all current non-tolerated pod will be removed; non-tolerated pod will not be scheduled.

✧ nodeSelector

- [pod.spec.nodeSelector](#) contains the labels (key:value pairs)
all the labels matches with the node => node is scheduled to this pod.

```
spec:
  nodeSelector:
    env: prod
```

✧ nodeAffinity

- [pod.spec.affinity.nodeAffinity](#) has to be updated.
nodeAffinity support complex queries like DoesNotExist, Exists, Gt, In, Lt, NotIn
 which was not possible in [nodeSelector](#)
- nodeAffinity is of 2 types:
 - [requiredDuringSchedulingIgnoredDuringExecution](#) [HARD rule]
 - [preferredDuringSchedulingIgnoredDuringExecution](#) [SOT rule (Scored basis)]

```
nodeAffinity
├─ preferredDuringSchedulingIgnoredDuringExecution []
│   └─ preference
│       └─ matchExpressions []
│           └─ matchFields []
│               └─ weight
└─ requiredDuringSchedulingIgnoredDuringExecution
    └─ nodeSelectorTerms []
        └─ matchExpressions []
            └─ matchFields []
```

- [~~its not match~~**Labels**, its **matchFields** (not limited to labels, can query any field like [metadata.name](#))]
- [matchExpressions](#): query1 **AND** query2 **AND** ...
- [matchFields](#): query1 **AND** query2 **AND** ...

```

matchExpressions:
- key: env
  operator: In
  values:
  - prod
- key: disk
  operator: In
  values:
  - ssd

matchFields:
- key: metadata.name
  operator: In
  values:
  - node01
  - node02
- key: metadata.resourceVersion
  operator: In
  values:
  - "353007"
  - "353008"

```

⚡ requiredDuringSchedulingIgnoredDuringExecution

```

requiredDuringSchedulingIgnoredDuringExecution
├─ nodeSelectorTerms [] (OR)
│   └─ Term 1
│       ├── matchExpressions [] (AND)
│       └─ matchFields [] (AND)
│           ⇒ Term 1 = (matchExpressions AND matchFields)
│
│   └─ Term 2
│       ├── matchExpressions [] (AND)
│       └─ matchFields [] (AND)
│           ⇒ Term 2 = (matchExpressions AND matchFields)

```

- ⚡ nodeSelectorTerms [] = term1 OR term2 OR ...
each term = (*matchExpressions* AND *matchFields*)
- ⚡ Node should satisfy any one *term* (means all queries of *matchExpressions* AND *matchFields* of that *term*)

⚡ preferredDuringSchedulingIgnoredDuringExecution

```

preferredDuringSchedulingIgnoredDuringExecution [] (scored list)
├─ Item 1
│   ├── preference
│   │   ├── matchExpressions [] (AND)
│   │   └─ matchFields [] (AND)
│   │       ⇒ match = (matchExpressions AND matchFields)
│   └─ weight
│
├─ Item 2
│   ├── preference
│   │   ├── matchExpressions [] (AND)
│   │   └─ matchFields [] (AND)
│   │       ⇒ match = (matchExpressions AND matchFields)
│   └─ weight

```



```
[
  { preference: (matchExpressions AND matchFields), weight },
  { ... },
  ...
]
```

Each item's preference:

weight = 'provided weight' if match else '0' (zero)

total score of node = sum of each item's weight

requiredDur....Execution: at least one should match

preferredDur...Execution: more matches, more score

Resource Requirements

Update field: `pod.spec.containers[*].resources`

`requests` -> required resources, else node will not be scheduled

[scheduling phase].

`limits` -> maximum utilization of resources in the node [running phase]

```
resources:
  requests:
    cpu: "500m"
    memory: "256Mi"
  limits:
    cpu: "1"
    memory: "512Mi"
```

cpu smallest unit: **1m** (1 mili CPU), **1** means one full CPU.

memory smallest unit: **1** (1 byte), use **Gi, Mi, Ki** (don't use G, M, K)

CPU maximum limit reached: throttled (slow down)

Memory maximum reached: container is killed

Best scenario: Request, No Limit

LimitRange

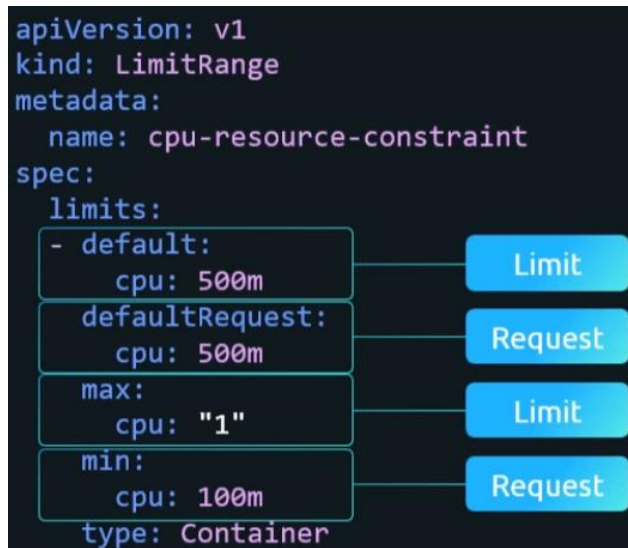
Inject default *request* and *limit* if not configured inside pods.

Multiple LimitRange can be configured in a namespace, but they should not conflict.

Ex: one says max cpu: 500m, another says min cpu: 600m

in this case, pod will not be created because of conflicting values.

- ☞ In real life: one LimitRange per namespace is maintained.



limits.default : limit

limits.defaultRequest: request

limits.max: request or limit should not more than it.

limits.min: request or limit should not less than it.

♪ ResourceQuota is for namespace level.

LimitRange is for Pod level.

♪ DaemonSet

same as ReplicaSet.

kind: DaemonSet

- ☞ It make sure one copy of the Pod runs on each node.

♪ Static Pods

- ☞ Kubelet on each node watches a directory, if Pod's yaml config file is present then it creates a pod.

It doesn't involve API Server.

- ☞ **systemctl cat kubelet** [see the *kubelet.service* file]

get the *config.yaml* file path and open that file.

One key will be there staticPodPath, that is the path to keep the Pod's yaml configs.

- ☞ Kube API Server knows about all the static pods, but you cannot update using **kubectrl edit** command.

The *yaml* config file needs to be changed to update the static pod.

♪ Pod creation internals

- ⌘ `kubectl apply -f my-pod.yaml` [pod creation request triggered]
- ⌘ API server validates, & creates Pod object. Update etcd about *PodSpec*.
- ⌘ Scheduler sees an unscheduled pod, assign a node.
- ⌘ *Kubelet* has a watch connection with API Server. Changes are streamed to kubelet from API server through that (its not webhook; kind of streaming type)
- ⌘ Kubelet sees *PodSpec*, pull it, command Container Runtime.
- ⌘ Container Runtime pull the required image, create the containers, assigns Container IDs.
- ⌘ Kubelet maps *Pod ID* -> *Container IDs* .
monitors containers (running/crashed/restarted)
- ⌘ Kubelet updates API Server about Pod's status (send details about containers like container IDs, pod status, ..etc)
- ⌘ API Server stores details in etcd.
[each few seconds, Kubelet updates API Server about the pod status, container details ..etc]
- ⌘ Container runtime knows about container only (container ID and all)
Kubelet knows about *PodSpec* (map it to container IDs received from container runtime)
API Server know keeps info that got from Kubelet only.
- ⌘ Each time we execute *kubectl edit* or *kubectl apply -f*, API Server sends *PodSpec* and the process continues.
- ⌘ In case of static pods, Kubelet sends a few extra fields
metadata.annotations."kubernetes.io/config.source": file
metadata.annotations."kubernetes.io/config.mirror": <hash>

🔗 Resources communication



- ⌘ All resources talk to API Server, API server talks to *etcd*.
All the communications happen via HTTP calls.
- ⌘ So, even if the resources are running in the node directly (or)

the resources are running inside pods, it doesn't matter.

Node to Pod will have some port mapping (in case of *kubeadm*, port mappings are same i.e. 6443:6443 ..etc)

- ⌘ The pods that is running resources, must be static. Because API Server shouldn't create or delete these.
- ⌘ **Kubelet** doesn't run inside any Pod. It is a **systemd** service.

✧ Multiple Schedulers

- ⌘ 3 files are involved:

kube-scheduler.yaml [kind: Pod , static pod that runs scheduler]

scheduler.conf [config file containing *cluster*, *user*, *context* details]

scheduler-config.yaml [kind: KubeSchedulerConfiguration; optional]

- ⌘ Either you mention the things in *pod.spec.containers[o].command* field without mentioning *scheduler-config.yaml* file [minimal configs like **--leader-elect=true**]

(or)

mention the *scheduler-config.yaml* file for proper configuration.

--config=/etc/kubernetes/scheduler-config.yaml [you can give any name]

```
spec:
  containers:
    - name: kube-scheduler
      image: k8s.gcr.io/kube-scheduler-amd64:v1.11.3
      command:
        - kube-scheduler
        - --address=127.0.0.1
        - --kubeconfig=/etc/kubernetes/scheduler.conf
        - --config=/etc/kubernetes/my-scheduler-config.yaml
```

- ⌘ To use this scheduler to schedule your pod, *pod.spec.schedulerName* field has to be updated.

schedulerName: my-custom-scheduler [pod.spec.schedulerName]

✧ Scheduler Profiles

- ⌘ Scheduler is a process (one pod)
Scheduler profile is a rule.
- ⌘ One scheduler can have multiple scheduler profiles.

- Scheduling has different stages called **execution point**.

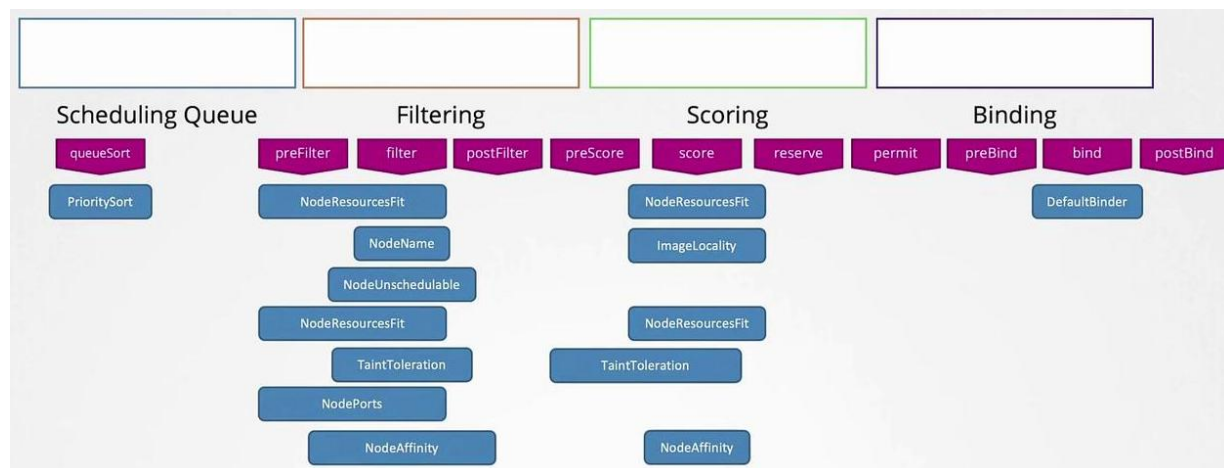
queueSort,

preFilter, filter, postFilter,

preScore, score, reserve,

permit, preBind, bind, postBind

- At each extension point, modules are bound.



- Scheduler profiles runs on different threads.

In each scheduler profile, you can decide which plugins has to be enabled or disabled.

```
apiVersion: kubescheduler.config.k8s.io/v1
kind: KubeSchedulerConfiguration
profiles:
- schedulerName: my-scheduler-2
  plugins:
    score:
      disabled:
      - name: TaintToleration
      enabled:
      - name: MyCustomPluginA
      - name: MyCustomPluginB
```

- KubeSchedulerConfiguration is not a Kubernetes object.

That's why this file's path is given in kube-scheduler.yaml

✧ Leader Selection

- `kubeschedulerconfiguration.leaderElect`: *true* (or) *false*

- Enable: one scheduler will be active at a time

Disable: multiple scheduler will be active at a time.

- ⌘ In case of race condition, Kubernetes handles it.
Only one scheduling can happen. No re-scheduling.

⌘

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